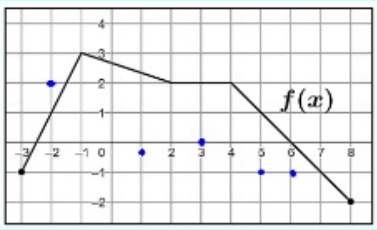
FUN-4.A9 Key features of functions and their derivatives can be identified and related to their graphical, numerical, and analytical representations.

FUN-4.A10 Graphical, numerical, and analytical information from f' and f'' can be used to predict and explain the behavior of f.

5-8 Sketching Graphs of Functions and Their Derivatives

Get Ready...

The graph of $f(x)$ is below. Find the following values of $f'(x)$ and plot these new points on the same grid as $f(x)$.



$f'(-2) = 2$

$f'(1) = \frac{-1}{3}$

$f'(3) = 0$

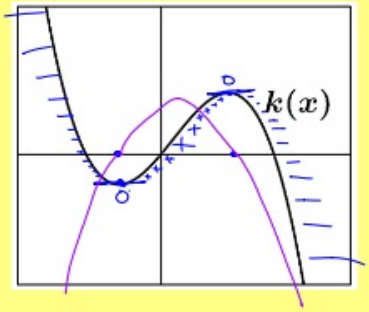
$f'(5) = -1$

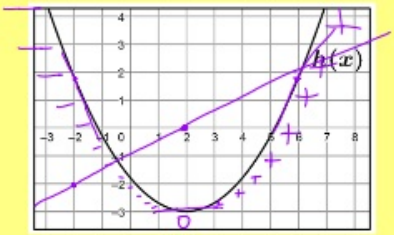
$f'(6) = -1$

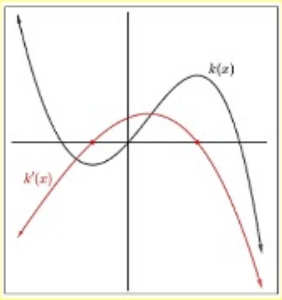
Important Reminder
The __slope__ of the graph of $f(x)$ will become the __y-value__ on the graph of $f'(x)$.

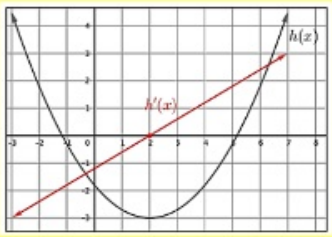
General Guidelines for sketching $f'(x)$ given $f(x)$
1.) Find any horizontal tangents on $f(x)$ — these are the __zeros (or x-intercepts)__ of $f'(x)$. 2.) If $f(x)$ is increasing, then $f'(x)$ is __positive__. (Above the __x-axis__) 3.) If $f(x)$ is decreasing, then $f'(x)$ is __negative__. (Below the __x-axis__)

Example 1: The graphs of $h(x)$ and $k(x)$ are below. Sketch the graphs of $h'(x)$ and $k'(x)$ on the appropriate axes.



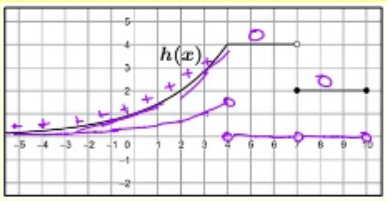
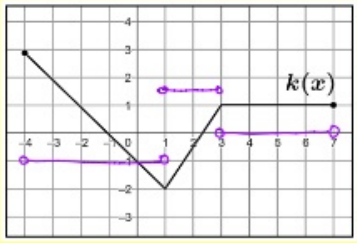
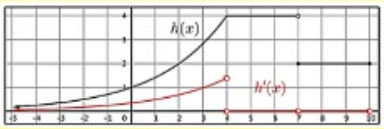
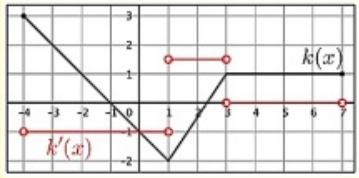






The graphs from Example 1 were polynomial functions, which made sketching a graph of the derivative relatively straightforward. However, we will encounter many functions that are NOT polynomials. These graphs may have some of the following interesting features:

1. No horizontal tangents
2. Vertical/Horizontal asymptotes
3. Points where the derivative does not exist
4. Discontinuous functions
5. Piece-wise functions

Important Notes
• If $f(x)$ is not continuous, has a sharp turn, or has a cusp, then $f(x)$ is NOT differentiable at that point. • If $f(x)$ is not differentiable at a point, then $f'(x)$ does not exist at that point. (This means that there will NOT be a point there on the graph of $f'(x)$ — this could be a hole, jump, or vertical asymptote)

Example 2: The graphs of several functions are given below. For each function, sketch the graph of its derivative.